



Bahria University, Islamabad
Department of Software Engineering

Numerical Analysis Lab
(Fall-2026)

Teacher: Engr. Sadaf Farhan

Student Name: Faisal khan

Enrollment: 01-131232-105

Lab Journal: 02

Date: 10-09-2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Error Accumulation

Task 1- Basic Arithmetic (AE & RE)

Write a program that reads two approximate floating-point numbers from the user and, based on the operation selected, computes the absolute error and the relative error for addition, subtraction, multiplication and division.

Your program should:

- Prompt for the two values and the operation.
- Compute the exact (true) result.
- Report $AE = |x' - x|$ and $RE = AE / |x'|$.
- Support +, -, ×, ÷.

Code:

```
# Task 1: Error Propagation with Automatic Error Detection

# 1. Prompt for inputs as strings to count decimal places automatically
str_x1 = input("Enter first number ")
str_x2 = input("Enter second number ")
op = input("Enter operation (+, -, *, /): ")

# Convert to float for math
x1 = float(str_x1)
x2 = float(str_x2)

# 2. Automatically calculate error based on decimal places (e = 0.5 * 10^-n)
dec1 = len(str_x1.split('.')[1]) if '.' in str_x1 else 0
dec2 = len(str_x2.split('.')[1]) if '.' in str_x2 else 0

e1 = 0.5 * (10 ** -dec1)
e2 = 0.5 * (10 ** -dec2)

print(f"\nCalculated error e1: {e1}")
print(f"Calculated error e2: {e2}")

# 3. Perform operation and propagate errors according to lecture rules
if op == "+":
    result = x1 + x2
    ae = e1 + e2 # Absolute errors add up for addition
    re = ae / abs(result) if result != 0 else 0

elif op == "-":
    result = x1 - x2
    ae = e1 + e2 # Absolute errors add up for subtraction
    re = ae / abs(result) if result != 0 else 0

elif op == "*":
    result = x1 * x2
    re1 = e1 / abs(x1) if x1 != 0 else 0
```

```
re2 = e2 / abs(x2) if x2 != 0 else 0
re = re1 + re2 # Relative errors add up for multiplication
ae = re * abs(result)

elif op == "/":
    if x2 != 0:
        result = x1 / x2
        re1 = e1 / abs(x1) if x1 != 0 else 0
        re2 = e2 / abs(x2) if x2 != 0 else 0
        re = re1 + re2 # Relative errors add up for division
        ae = re * abs(result)
    else:
        result = 0
        ae = 0
        re = 0
        print("Error: Division by zero!")
else:
    result = 0
    ae = 0
    re = 0
    print("Invalid operation!")

# 4. Calculate the final range interval where the true value lies
min_val = result - ae
max_val = result + ae

print("\n--- Final Results ---")
print(f"Result (z): {result}")
print(f"Absolute Error (AE): {ae}")
print(f"Relative Error (RE): {re}")
print(f"Range: {min_val} <= true value <= {max_val}")
```

Output:

```
Enter first number 4.0643
Enter second number 0.37487
Enter operation (+, -, *, /): *

Calculated error e1: 5e-05
Calculated error e2: 5e-06

--- Final Results ---
Result (z): 1.523584141
Absolute Error (AE): 3.9065e-05
Relative Error (RE): 2.564019862687715e-05
Range: 1.523545076 <= true value <= 1.523623206

c:\Users\hp\Desktop\NA-Lab-2>
```

Task 2 - Error in the Area of a Circle

Write a program that asks the user to enter the radius r and its possible error Δr . The program should then compute the range in which the true area of the circle lies.

- Read r and Δr from the user.
- Area $A = \pi r^2$
- Print the interval $A_{\min} \leq \text{true area} \leq A_{\max}$

Code:

```
# Task 2: Error in the Area of a Circle
import math

# Read radius (r) and its possible error (Ar) from the user
r = float(input("Enter radius (r): "))
delta_r = float(input("Enter radius error (Ar): "))

# Compute minimum and maximum possible radius values
r_min = r - delta_r
r_max = r + delta_r

# Compute minimum and maximum area using Area = pi * r^2
amin = math.pi * (r_min ** 2)
amax = math.pi * (r_max ** 2)

# Print the final interval
print(f"{amin} <= true area <= {amax}")
```

Output:

```
Enter radius (r): 5
Enter radius error (Ar): 0.266
70.40546641099337 <= true area <= 87.11873932809108

C:\Users\hp\Desktop\NA-Lab-2>
```

Thank you